

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DARES SALAAM REGION

FORM FOUR - MOCK EXAMINATION - 2025

PHYSICS

GEPAM science hub

WhatsApp 0748978398

CODE:031

TIME:3:Hours

Monday 24th July, 2025 A.M

INSTRUCTIONS

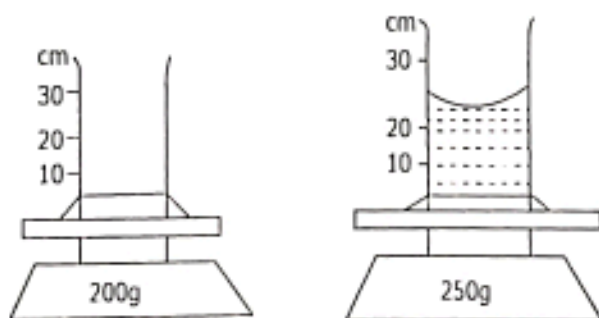
1. This paper consists of three (03) sections A, B and C with total of eleven (11) questions
2. Answer **all** questions in section A and B and only two (02) question from section C
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. Communication devices and unauthorized materials are **not** allowed in the examination room.
5. Non – programmable calculators and mathematics table may be used.
6. Write your **Examination Number** on every page of your answer booklet(s)
7. Where necessary the following number on every page may be used.
 - i. Acceleration due to gravity $g = 10\text{m/s}^2$
 - ii. Linear expansivity of aluminum $= 0.00003^\circ\text{C}$
 - iii. Density of water $= 1\text{g/cm}^3$

This paper consists of 6 printed pages

SECTION A (16 Marks)

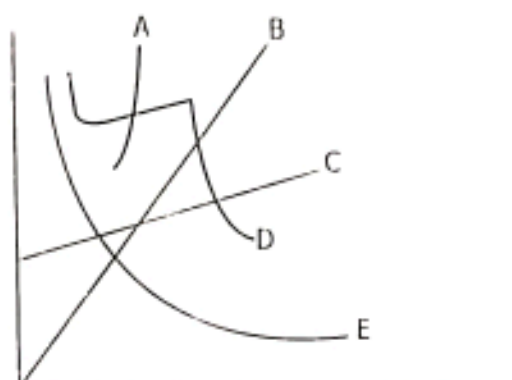
For each of the following item (i – x), choose the correct answer among the given alternatives and write its letter beside the item number

- i. Which of the following best describes why the knowledge of physics is necessary to understand all other Science
 - A. Physics explains how energy passes from one object to another
 - B. Physics explains how gravity works
 - C. Physics explains the motion of objects that can be seen with the naked eye
 - D. Physics explains the fundamental aspect of the universe
 - E. Physics describes the life of an organism
- ii. In a light experiment, observing an object through a certain material showed that less light was transmitted and the image was distorted. Which type of material was used.
 - A. A translucent material
 - B. An opaque material
 - C. A luminous material
 - D. A transparent material
 - E. A non-luminous material
- iii. Which metals become strongly magnetized when subjected to a magnetic field
 - A. Nickel and copper
 - B. Zinc and Aluminium
 - C. Cobalt and Iron
 - D. Aluminium and lead
 - E. Iron and Zinc
- iv. A stone is thrown horizontally from the top of a building at a speed of 12m/s. How long does it take to fall 45m vertically? (Assume negligible air resistance)
 - A. 1.0 sec
 - B. 5.0 sec
 - C. 3.03 sec
 - D. 8.0 sec
 - E. 2.14 sec
- v. In physics test students were asked to mention the examples of friction. The response of one candidate was listed below.
 1. Tension in a string
 2. Air resistance
 3. Weight
 Which among the alternatives shows the correct examples of friction?
 - A. 1 only
 - B. 1 and 3
 - C. 2 only
 - D. 2 and 3
 - E. 3 only
- vi. What is the density of the liquid in the figure below?



- A. 2.0g/cm^3
- B. 8.0g/cm^3
- C. 10.0g/cm^3
- D. 0.5g/cm^3
- E. 0.1g/cm^3

- vii. Ten best students from Murubona International school were taken to China and they visited one of the lakes there and they found that Pond skaters were walking on the surface of that lake. This is because of the
- Up thrust of the water
 - Density of that water
 - Surface tension of that water
 - Capillarity of that water
 - Force of that water
- viii. The diagram below shows five graphs of quantities in heat A,B,C,D and E



Which of these best represent the variation of the volume of a fixed mass of gas with the temperature in $^{\circ}\text{C}$ when the pressure is being constant.

- ix. When Silicon element is doped with phosphorus atom it produces
- A P- type semiconductor
 - A P – junction diode
 - An intrinsic semiconductor
 - A bipolar semiconductor
 - An n - type semiconductor
- x. Two waves come together and interact to form a new wave of smaller amplitude. What is the type of wave interference?
- Constructive interface
 - Destructive interface
 - Reflective interference
 - Positive interference
 - Negative interference
2. Match the properties of radiations in **List A** with the corresponding radiations in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided. One response can be used more than one

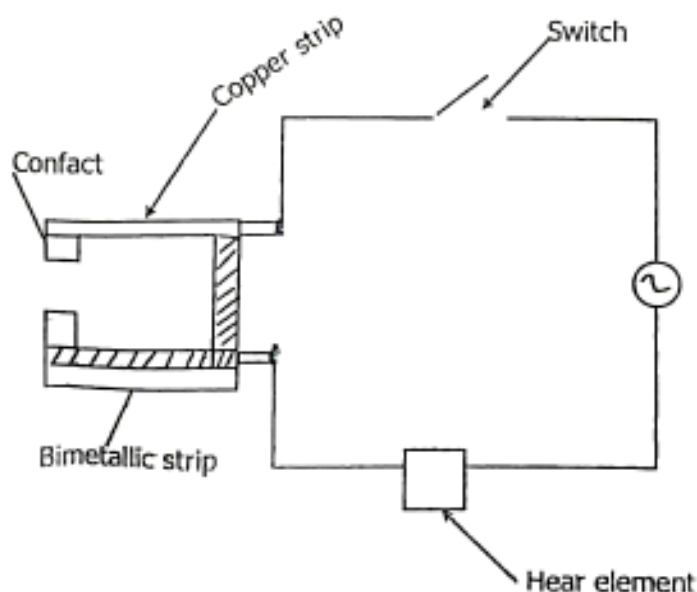
LIST A	LIST B
i. Has weak - moderate ionizing power ii. Is deflected towards south pole of the magnet iii. Has high penetrating power but stopped by lead sheet iv. Has the least penetrating power but stopped by a sheet of paper v. Has a speed up to 10% of the speed of light in vacuum vi. Has very weak-low ionizing power	

SECTION B (54 Marks)

3. (a) Explain the appearance of a red tie with blue spots when observed in
- Red light
 - Green light

(04 marks)

- (b) A telescope of 50m diameter with a reflector of focal length 18.0m is used to focus the image of the sun. Using the distance of the sun from the earth and diameter of the sun as $1.5 \times 10^{11}\text{m}$ and $1.4 \times 10^9\text{m}$ respectively, calculate the
- position of the image of the sun
 - Diameter of the image of the sun
- (05marks)**
4. (a) Form one students from Mzumbe Secondary School visited Kilimanjaro Mountain. When climbing the mountain to high altitude, one of the students got the problem of nose bleeding
- Comment on why student got such problem
 - Why astronauts wear space suits
- (04marks)**
- (b) A uniform half meter rule is balanced at 15cm mark when a load of mass 40g is hanging at zero mark
- Draw a sketched diagram to show the system
 - Determine the mass of the half meter rule
- (02 marks)**
(03marks)
5. (a) A pulley system with efficiency of 75% is made up to four pulleys. Sketch a well labelled diagram of this machine and then determine the effort required to raise two bags of maize each weighing 100kg
- (05 marks)**
- (b) i. Sketch the diagram of a body which starts from rest and accelerates uniformly for sometimes to a constant velocity for a certain period of time before decelerating uniformly to a stop
- (02marks)**
- ii. A box of mass 60kg is raised vertically with a uniform acceleration "a" when a force of 900N is acting in a rope. Calculate its uniform acceleration. **(02marks)**
6. (a) Briefly explain how does the transfer of thermal energy occur in metal spoon that has been placed in a pot of boiling water
- (04marks)**
- (b) The figure below shows a circuit diagram for controlling the temperature of a room.



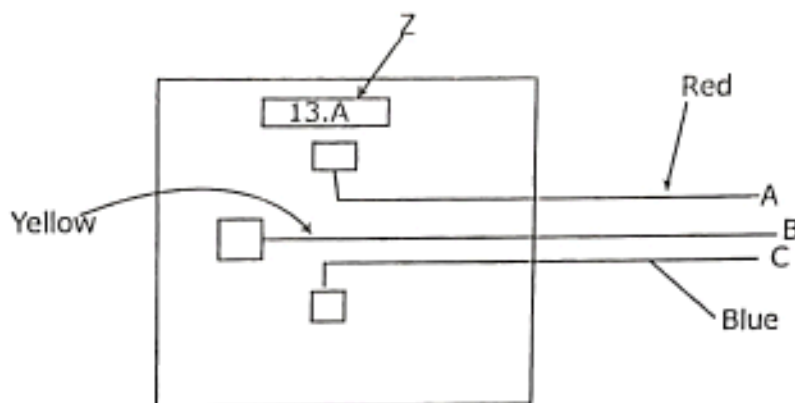
- Explain the friction of the bimetallic strip
 - Describe how the circuit controls the temperature when the switch is closed
- (02marks)**
(03 marks)

- (a) Draw a well labelled diagram of a cathode ray oscilloscope (05marks)
 (b) Water ripples are caused to travel across the surface of a shallow tank by means of a suitable straight vibrator. The distance between successive crests and troughs is 1.5cm and the wave travels 25.2cm in 1.2 second calculate
 i. The velocity of the waves (02marks)
 ii. the frequency of the vibrator (02marks)
8. (a) Explain five (05) measure which the government of Tanzania take to reduce the emission of greenhouse gases into the atmosphere (05marks)
 (b) Explain how tides occur (04marks)

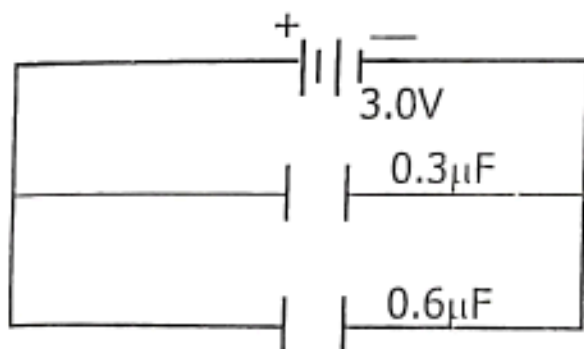
SECTION C (30 Marks)

Answer only **two (02)** questions

9. (a) Given the three pin plug below with component Z in it



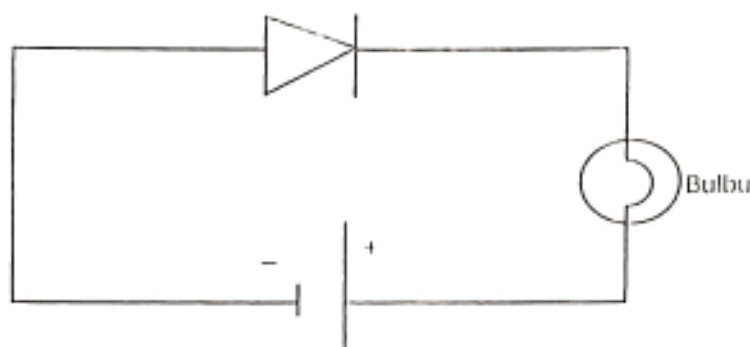
- What are the names of the cable A, B, and C
 - What is the name of component Z
 - What will happen to component Z if 14A current flowing through it?
 - What is the purpose of the component Z in the plug? (4 ½ Marks)
- (b) A bulb with a resistance of 5Ω is connected to an 8V battery. If the internal resistance of the battery is neglected.
 i. How much current will flow through the bulb?
 ii. How much light and heat energy does the lamp produce in one hour (4 ½ Marks)
- (c) i. Determine the effective capacitance of the circuit show below
 ii. How much charge is stored



(03marks)

10. (a) During project preparation a group of form four students visited a radiology department of a near by hospital. Their aim was to observe an electromagnetic wave 'A' which is used in hospital to obtain diagnostic information and treatment of different diseases. Assume that you are a head of that department.
- Name the type of electromagnetic wave 'A' (01 mark)
 - Provide any four properties of the wave 'A' (04 marks)
 - Briefly explain how wave 'A' is produced (02 marks)

- (b) A physics teacher provided diode bulb, cell and connecting wires and ask his students to connect the components to form a circuit. One of his students connected the circuits as shown below



- What type of biasing is that ?
 - Will the bulb work or not ? Explain why (03 marks)
- (c) One end of an N-type doped semiconductor is heated
- What is expected between the heated end and the cooled end (02 1/2 marks)
 - Identify the negative end of the two ends (02 1/2 marks)
11. (a) A low- voltage out door-lighting system uses a transformer to step down the 240V house hold voltage to 24V. The lighting system has 6 lamps with a total resistance of 9.6Ω (0 1/2 mark)
- What is the current in the secondary coil ? (03 marks)
 - What is the current in the primary coil ? (03 marks)
- (b) For each of the following electromagnetic wave describes a process by which it is produced.
- Visible light
 - Microwaves
 - Radiowaves (4 1/2 marks)
(@ 1 1/2)
- (c) A form four student was given the following musical instruments and asked to put them into their proper groups. Guitar, xylophone, trumpet, drum, flute, and violin. Name the groups and put the instruments into their proper groups. (4 1/2 marks)

DAR ES SALAAM REGION
FORM FOUR MOCK EXAMINATION
PHYSICS PAPER 1
MARKING SCHEME

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SECTION A (16 Marks)

I	II	III	IV	V	VI	VII	VIII	IX	X
D	A	C	C	C	A	C	C	E	B

@ 1 Mark = 10 marks

2 Matching Items

I	II	III	IV	V	VI
B	A	C	A	B	C

@ 1 mark = 06 Marks

SECTION B (54 Marks)

- 3 (a) (i) Red tie with magenta spots - - - (02)
 (ii) Yellow tie with cyan spots - - - (02)

(b) Data

$$f = 18.0 \text{ m}$$

$$u = 1.5 \times 10^{11} \text{ m}$$

$$d = 1.4 \times 10^9 \text{ m}$$

Since the object (the sun) is very far, nearly $1.5 \times 10^{11} \text{ m}$. It can be assumed to be at infinity.

$$v = f = 18.0 \text{ m} \quad \text{--- (01 mark)}$$

- (i) $\therefore$ The image is 18.0 m from the reflector

$$\text{Magnification, } m = \frac{\text{Image distance } v}{\text{Object distance } u}$$

3 (b)

$$= \frac{18.0 \text{ m}}{1.5 \times 10^{11} \text{ m}} \quad \dots \dots (01 \text{ Mark})$$

$$= 1.2 \times 10^{-10} \quad \dots \dots (00\frac{1}{2} \text{ Mark})$$

(ii) Then, Magnification, $m = \frac{\text{Image diameter } d'}{\text{Object diameter } d}$
(01 Mark)

$$1.2 \times 10^{-10} = \frac{d'}{d}$$

$$d' = 1.2 \times 10^{-10} \times 1.4 \times 10^9 \text{ m}$$

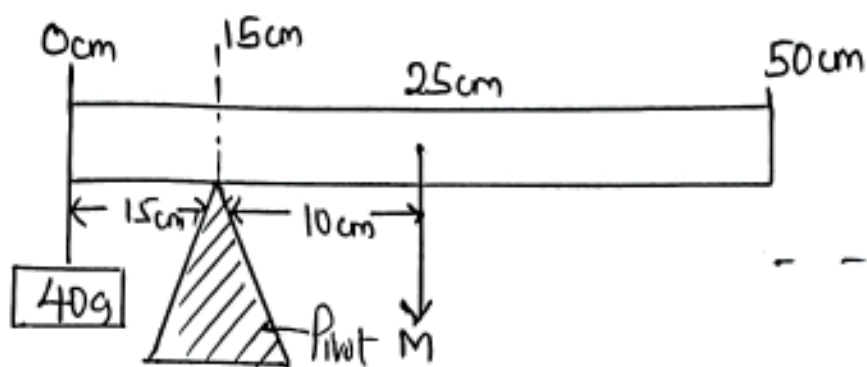
$$\therefore d' = 0.168 \text{ m} \quad (00\frac{1}{2} \text{ Mark})$$

$\therefore$ The Image diameter is 16.8 cm
(01 Mark)

4 (a) (i) The student got problem of nose bleeding due to high blood pressure at high altitude compared to atmospheric pressure
(02 Marks)

(ii) Astronauts wear space suits in order to maintain the pressure.
(02 Marks)

4 (b)(i) Diagram



--- (02 marks)

(ii) To find mass of the ruler
From
Principle of moments.

clockwise moment = Anticlockwise moment (0.5 mark)

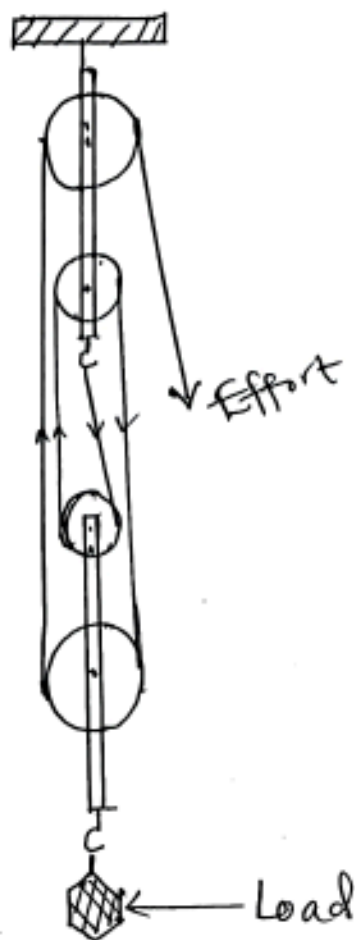
$$m \times 10\text{cm} = 40\text{g} \times 15\text{cm} \quad \text{--- (01 mark)}$$

$$m = \frac{40\text{g} \times 15\text{cm}}{10\text{cm}}$$

$$m = 60\text{g} \quad \text{--- (01 mark)}$$

$\therefore$ The mass of the half metre ruler is 60g
(0.5 mark)

5 (a) Diagram



(02 Marks)

Data

$V.R = \text{Number of pulleys} = 4$

$e = 75\%$

$L = 100 \text{ kg} \times 2$

$E = ?$

$e = \frac{M.A}{V.R} \times 100\%$. . . (00 1/2 mark)

$M.A = \frac{L}{E}$. . . (00 1/2 mark)

$e = \frac{\text{Load}}{E \times V.R} \times 100\%$. . . (00 1/2 Marks)

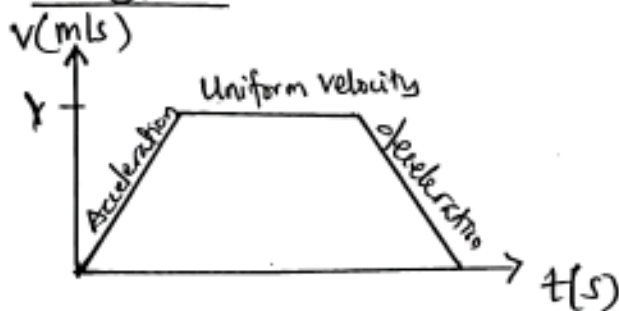
5 (a) $75\% = \frac{200\text{kg}}{E \times 4} \times 100\%$ - - - (00½ mark)

Effort = $\frac{200\text{kg}}{0.75 \times 4}$ - - - (00½ mark)

= 66.67kg

∴ The effort required to raise the mass of 200kg is 666.7N - - (00½ mark)

(b) (i) Diagram.



(ii) Data

Mass(m) = 60kg

Force(F) = 900N

form

Newton's second law of motion.

$F = ma$

$a = \frac{F}{m}$

= $\frac{900\text{N}}{60\text{kg}}$

$a = 15\text{N/kg}$

$a = 15\text{m/s}^2$

∴ The uniform acceleration of the box is 15m/s^2

(a) The transfer of thermal energy from the boiling water to the metal spoon occurs primarily through conduction. The high energy water molecules collide with the spoon's surface, transferring kinetic energy to the free electrons in the metal.

These electrons highly mobile within the metal's structure, distribute their energy throughout the spoon via collisions, leading to a rise in the spoon's temperature.

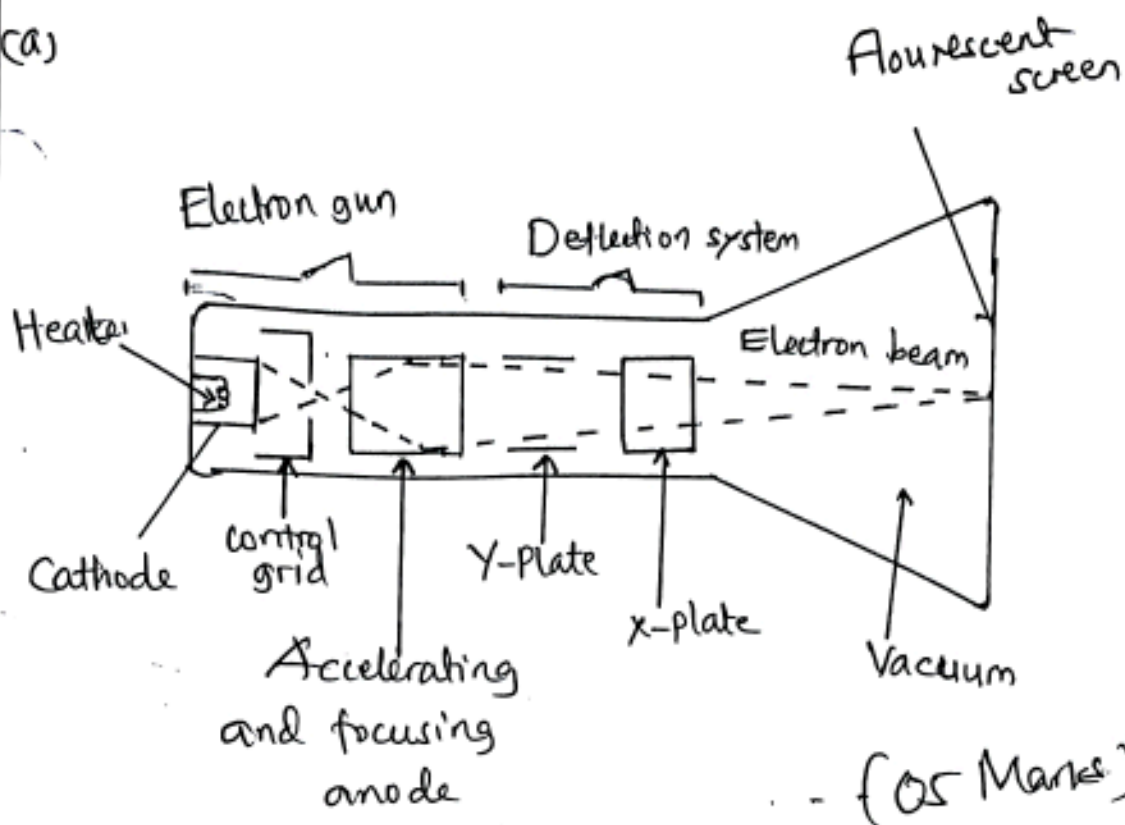
(04 Marks)

(b) (i) It is used to control flowing of current in the circuit --- (02 Marks)

(ii) When the switch is closed, electricity flows through the circuit, including the heater element. The heater element produces heat, raising the room's temperature. As the room warms up, the bimetallic strip also heats up. The strip bends due to the different expansion rates of the metals. When the temperature reaches a certain point, the strip bends enough to break the contact, stopping the flow of electricity and turning off the heater. As the room cools, the bimetallic strip cools, bends back, and closes the circuit again, restarting the heating process. This cycle continues, maintaining a relatively constant temperature.

- . (03 Marks)

7 (a)



7 (b) Data

Length of crest (L) = 1.5 cmDistance (d) = 25.2 cmtime (t) = 1.2 s.(i) To calculate the velocity of the wave (v).

From

$$d = v \times t$$

$$v = \frac{d}{t}$$

$$= \frac{25.2 \text{ cm}}{1.2 \text{ s}} \quad \dots \quad (0.5 \text{ Mark})$$

$$v = 21 \text{ cm/s} \quad \dots \quad (0.1 \text{ Mark})$$

$$v = 21 \text{ cm/s}$$

 $\dots (0.5 \text{ Mark})$

$\therefore$ The velocity of the wave is 21 cm/s.

7 (b)(ii) To calculate the frequency of the vibrator
from

$$v = f\lambda$$

$$f = \frac{v}{\lambda}, \text{ but } \lambda = 2L \dots (00\frac{1}{2} \text{ Mark})$$

$$f = \frac{v}{2L} \dots (00\frac{1}{2} \text{ Mark})$$

$$= \frac{21 \text{ cm/s}}{2 \times 1.5 \text{ cm}} \dots (00\frac{1}{2} \text{ Mark})$$

$$\therefore f = 7 \text{ Hz} \dots (00\frac{1}{2} \text{ Mark})$$

$\therefore$ The frequency of the vibrator is 7 Hz

8 (a) Measures.

- (i) Afforestation (Planting of trees)
- (ii) Recycling (Encourage recycling of waste products)
- (iii) Government adopt carbon emission policy
- (iv) Encourage farmers to use organic farming
which is free from fertilizers that add
pollutant.
- (v) Use clean alternative sources of energy
(wind, biomass and geothermal)

(@ 1 = 5 marks)

(b) Ocean tides occur by interaction between the sun, the earth, the moon and the ocean waves results in ocean tides.

(4 marks)

SECTION C (30 marks)

- 9 (a) (i) $\left. \begin{array}{l} A - \text{Live} \\ B - \text{Earth} \\ C - \text{Neutral} \end{array} \right\} (0\frac{1}{2} \text{ Marks})$

(ii) Fuse - - - (01 mark)

(iii) It will blow off - - (01 mark)

(iv) To control the amount of flowing current
(01 mark)

(b) Data

$$R = 5\Omega$$

$$V = 8V$$

$$t = 1 \times 60 \times 60s = 3600s$$

(i) Current flowing
from
Ohm's Law

$$V = IR \quad \dots \dots (0\frac{1}{2} \text{ Mark})$$

$$I = \frac{V}{R}$$

$$= \frac{8V}{5\Omega} \quad \dots \dots (01 \text{ Mark})$$

$$I = 1.6A$$

- The current flowing is $1.6A$ (00 $\frac{1}{2}$ mark)

(b)(ii) To find energy

$$E = \frac{V^2}{R} \times t \quad \dots \dots (01 \text{ mark})$$

$$E = \frac{8 \times 8 \times 3600}{5} \text{ J} \quad \dots (01 \text{ mark})$$

$$E = 46080 \text{ J} \quad \dots (00\frac{1}{2} \text{ mark})$$

∴ Heat and light energy produced was
46080 J.

9 (c) Data

(i) For parallel capacitor.

$$C_T = C_1 + C_2 \quad \dots \dots (01 \text{ mark})$$

$$= 0.3 + 0.6 \quad \dots \dots (01 \text{ mark})$$

$$C_T = 0.9 \mu\text{F} \quad \dots (01 \text{ mark})$$

(ii) Charged stored

$$q = C V \quad \dots \dots (01 \text{ mark})$$

$$= 0.9 \times 10^{-6} \times 3.0 \text{ V} \quad \dots (01 \text{ mark})$$

$$q = 2.7 \times 10^{-6} \text{ C} \quad \dots (01 \text{ mark})$$

10 (a) (i) X-rays - - - (01 mark)

(ii) Properties.

- They travel in straight lines with the speed of light in vacuum, $3 \times 10^8 \text{ m/s}$
- They can penetrate various objects.
- They are not affected by electric or magnetic fields.
- They cause fluorescence upon striking certain types of crystals
- They can ionize gases causing the gases to conduct electricity

(Any four) @ 01 = 04 marks

(ii) X-rays are produced when fast moving electrons strike a metal target and lose their energy.

(02 Marks)

(b) (i) Reverse biasing - - (01 mark)

(ii) The bulb will not work since no or few electrons/current allowed to flow through the circuit.

(02 Marks)

- (c) (i) Potential difference will be developed between the heated and the cooled ends of the semiconductor. (02.5 marks)
- (ii) The colder end will be negative (02½ marks)

11 (a) Data

Primary voltage (V_p) = 240V

Secondary voltage (V_s) = 24V

Output resistance (R) = 9.6Ω

(i) To find current in secondary coil (I_s)
from

$$V_s = I_s R$$

$$I_s = \frac{V_s}{R} \quad \dots (01 \text{ mark})$$

$$= \frac{24V}{9.6\Omega} \quad \dots (01 \text{ mark})$$

$$I_s = 2.5A$$

∴ The current in the secondary coil is 2.5A
(00 1 Mark)

11 (ii) To find current in primary coil (I_p)
from
Ideal transformer.

$$I_s V_s = I_p V_p$$

$$I_p = \frac{I_s V_s}{V_p} \dots (01 \text{ mark})$$

$$= \frac{2.5 \text{ A} \times 24 \text{ V}}{240 \text{ V}} \dots (01 \text{ mark})$$

$$I_p = 0.25 \text{ A}$$

- The current flowing in the primary coil is 0.25 A - - (00.5 mark)

(b) (i) Visible light:- By electrons transition within an atom - - (01½ mark)

(ii) Microwaves:- Artificially:- devices such as circuits, transmission towers, RADAR, magnetron
(01½ mark)

(iii) Radio waves:- Charged particles undergoing acceleration such as time-varying electric current
(01½ mark)

(c) String Instruments:- Guitar, Violin ($0\frac{1}{2}$ mark)

Percussion Instruments: Drum, Xylophone ($0\frac{1}{2}$ mark)

Wind Instruments:- Trumpets, flute ($0\frac{1}{2}$ mark)